

ISyE 3770 HW10

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Problem 10.2.6

Part A

We will utilize Welch's t-test.

We begin by calculating the standard error:

$$SE = \sqrt{\frac{12^2}{10} + \frac{22^2}{16}} = \sqrt{44.65} = 6.682$$

Then, we determine the test statistic:

$$t = \frac{321 - 290}{6.682} = 4.64$$

Now, we use R to speed up calculations:

```
n1 <- 10
s1 <- 12
n2 <- 16
s2 <- 22

numerator <- ( (s1^2 / n1) + (s2^2 / n2) )^2

denominator <- ( ( (s1^2 / n1)^2 ) / (n1 - 1) ) + ( ( (s2^2 / n2)^2 ) / (n2 - 1) )

df <- numerator / denominator

df

## [1] 23.72113

t_value <- 4.64
p_val <- 1 - pt(t_value, df)
p_val

## [1] 5.324027e-05
```

Therefore, since we have a p-value of $0.00005 > 0.05 = \alpha$, we reject the null hypothesis. Therefore, we find there is strong evidence to support the claim that Supplier 2 provides more gears with higher mean impact strength.

Part B

We again calculate the test statistic:

$$t = \frac{(321 - 290) - 25}{6.682} = 0.898$$

```
t_value <- 0.898
df <- 23.72113
p_value <- 1 - pt(t_value, df)
p_value
```

```
## [1] 0.1891051
```

Since the p-value of $0.1891 > 0.05 = \alpha$, we fail to reject the null hypothesis. Therefore, we find the data does not support the mean impact strength of gears from Supplier 2 is at least 25 ft-lbs higher than that of Supplier 1.

Part C

We construct a 95% CI:

```
df <- 23.72113
alpha <- 0.05
t_star <- qt(1 - alpha/2, df)
t_star
```

```
## [1] 2.065183
```

We then calculate the margin of error:

$$ME = t * (SE) = 2.065(6.682) = 13.79833$$

Therefore, we observe a CI of:

$$(321 - 290) \pm 12.79 = 31 \pm 13.79$$

This gives us a lower limit of 17.21 and an upper limit of 44.79. Therefore, we find the 95% CI to be (17.21, 44.79).

Since the entire interval is > 0 , it supports the conclusion from Part A. Additionally, since the lower limit is < 25 , it supports the conclusion in Part B.

The CI suggests the true difference is between 17.21 and 44.79 ft-lbs.